Hashing Technique – C Program

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#include <stdio.h>
#define SIZE 10

int main()
{
    int table[SIZE], key, i, index, choice;

    for (i = 0; i < SIZE; i++)
        table[i] = -1;

    do
    {
        printf("\n1.Insert  2.Search  3.Display  4.Exit");
        printf("\nEnter choice: ");
        scanf("%d", &choice);

        switch (choice)
        {
            case 1:
                printf("Enter value: ");
                scanf("%d", &key);

                index = key % SIZE;
                i = 0;

                while (i < SIZE &&
                       table[(index + i) % SIZE] != -1)
                    i++;

                if (i == SIZE)
                    printf("Hash table is full");
                else
                {
                    table[(index + i) % SIZE] = key;
                    printf("Inserted successfully");
                }
                break;

            case 2:
                printf("Enter value to search: ");
                scanf("%d", &key);

                index = key % SIZE;
                i = 0;

                while (i < SIZE &&
                       table[(index + i) % SIZE] != -1 &&
                       table[(index + i) % SIZE] != key)
                    i++;

                if (i < SIZE &&
                    table[(index + i) % SIZE] == key)
                    printf("Element found at index %d",
                           (index + i) % SIZE);
                else
                    printf("Element not found");
                break;

            case 3:
                printf("\nIndex\tValue\n");
                for (i = 0; i < SIZE; i++)
                {
                    if (table[i] == -1)
                        printf("%d\t-\n", i);
                    else
                        printf("%d\t%d\n", i, table[i]);
                }
                break;

            case 4:
                printf("Exit");
```

```
                break;

            default:
                printf("Invalid choice");
        }
    } while (choice != 4);

    return 0;
}
```